

Advanced Counting

- 1 A college has 1105 students. This semester 320 students are taking a math course, 405 of the students are taking a history class, and 250 of the students taking a math class are not enrolled in a history class.
 - (a) Construct a Venn diagram, and write the appropriate numbers in the four regions
 - (b) How many of the students are taking math and history?
 - (c) How many students are taking history but not math?
 - (d) How many students are taking math or history?
 - (e) How many students are taking neither math nor history?

- 2 *Survivor*, *CSI*, and *ER* all belong to the Thursday night prime time lineup. A group of 600 college students were surveyed. 350 students watch *Survivor*, 280 watch *CSI*, and 320 watch *ER*. 50 students watch all three shows. 190 students watch both *Survivor* and *ER*. 180 students watch both *Survivor* and *CSI*. 50 students watch only *ER*.
 - (a) Construct a Venn diagram, and write the appropriate numbers in the eight regions.
 - (b) How many students watch only *Survivor*?
 - (c) How many students watch *CSI* or *Survivor*?
 - (d) How many students watch *CSI* or *ER*?
 - (e) How many students watch only *CSI*?

- 3 A certain model of pickup truck is available in five exterior colors, three interior colors, and three interior styles. In addition, the transmission can be either manual or automatic, and the truck can have either two-wheel or four-wheel drive. How many different versions of the pickup truck can be ordered?

- 4 The $\Delta\Delta\Delta$ fraternity must elect a president, vice president, and treasurer from its 30 members. How many ways can this be done?

- 5 The local branch of a national business chain has 27 employees. If they all shake each others hand at the company picnic, at least how many handshakes occurred?

- 6 Using a standard deck of 52 cards,
 - (a) how many ways can five cards be drawn?
 - (b) how many five card hands can be dealt?

- 7 A three-person committee must be chosen from a group of ten men and twelve women. How many committees are possible if it must consist of the following?
- (a) Three men.
 - (b) Three women.
 - (c) One woman and two men.
 - (d) One man and two women, and one of these is selected as a spokesperson of the committee.
- 8 Each student at State University has a student ID number consisting of five digits (the first digit is nonzero, and the digits can be repeated) followed by two of the letters A , B , C , and D (letters cannot be repeated). How many different student numbers are possible?
- 9 How many ways are there to place 6 identical objects into 3 different bins?
- 10 How many ways can we add three non-negative integers such that they sum to 9? (order matters)
- 11 Which of the following functions maps a 11-to-1 correspondence from the set $1, 2, 3, \dots, 44$ to the set $1, 2, \dots, 11$?
- A. $f(x) = \binom{x}{3}$ B. $f(x) = \lfloor \frac{x}{11} \rfloor$ C. $f(x) = \lceil \frac{x}{11} \rceil$ D. $f(x) = \lfloor \frac{x}{11} \rfloor$
- 12 Which of the following functions maps a 11-to-1 correspondence from $\{1, 2, 3, \dots, 44\}$ to $\{0, 1, \dots, 10\}$?
- A. $f(x) = \binom{x}{3}$ B. $f(x) = \lfloor \frac{x}{11} \rfloor$ C. $f(x) = \lceil \frac{x}{11} \rceil$ D. $f(x) = \lfloor \frac{x}{11} \rfloor$
- 13 What is the coefficient of x^6y^4 in $(x - 2y)^{10}$?
- 14 What is the value of $\binom{10}{0} - \binom{10}{1} + \binom{10}{2} - \dots + \binom{10}{10}$?
- 15 Suppose that $B = \{0, 1, 2, \dots, 8\}$. If $f : A \rightarrow B$ is 3-to-1 mapping of A onto B , what is $|A|$?

Discrete Probability

- 1 A card is dealt from a well-shuffled, standard deck of 52 cards. Find the probability of being dealt each of the following.
 - (a) A red card
 - (b) A queen
 - (c) A club
 - (d) The queen of clubs
 - (e) A queen or a club
 - (f) Not a queen

- 2 Three fair coins are tossed. Find each of the following
 - (a) The sample space
 - (b) The event, E_1 , that exactly two are tails.
 - (c) The event, E_2 , that at least two are tails.
 - (d) The probability of E_1
 - (e) The probability and odds of E_2

- 3 Five cards are dealt (without replacement) from a well-shuffled 52 card standard deck. Find the probability of the following:
 - (a) being dealt 5 spades.
 - (b) being dealt 4 spades and 1 heart.
 - (c) being dealt all the same suit.
 - (d) not being dealt all the same suit

- 4 A drawer has 7 socks. 4 socks are black and 3 are white socks. John randomly pulls out 4 socks. Find the probability of the following:
 - (a) all 4 socks are black.
 - (b) exactly 2 are white.
 - (c) at least 3 are white.
 - (d) at most 2 are black.

- 5 In a certain game, a player bets \$5. There is a 25% chance that the player wins \$2, a 25% chance the player wins \$7, and a 50% chance the player loses \$5. What is the expected outcome of this game? Round to the nearest cent.
- 6 WGU is holding a raffle to raise money for a new internet service. The tickets cost \$10 each. The value and number of each prize to be given away is listed in the provided table. Find the expected value of a ticket under the given circumstances.

Prize	Value	#
Vacation	\$3,000	1
Stereo	\$500	5
Books	\$100	10
T-shirt	\$20	100

- (a) 200 tickets
- (b) 1000 tickets
- 7 A fair six-sided dice is rolled 5 times. Find the probability of the following:
- (a) All 2's are rolled.
- (b) Four 2's are rolled and the 5th roll is a 1.
- (c) Not all 2's are rolled.

Baye's Theorem

- 1 Bag #1 contains 10 blue marbles. Bag #2 appears identical but contains 5 red marbles and 5 blue marbles. Suppose you randomly choose a bag and then randomly pick a blue marble. What is the probability that you choose bag #2?

- 2 The zombie apocalypse has come. However, it's not as bad as the movies. Only %1 of the population will get infected. 90% of the people who are infected with this disease will test positive, but %5 of the people tested will have false positives (meaning the test shows positive but they are **not** infected). What is the probability that a person is infected if the test shows positive? Approximate your answer to the 4th decimal place.